

Linear Algebra and PCA

An eigenvector of matrix A satisfies $Av = ev$ where e is the eigenvalue. Eigendecomposition factorizes a matrix into eigenvectors and eigenvalues. PCA uses eigenvectors of the covariance matrix to reduce dimensionality. The first principal component captures maximum variance. SVD generalizes eigendecomposition to non-square matrices. Dot product and matrix multiplication are fundamental operations.